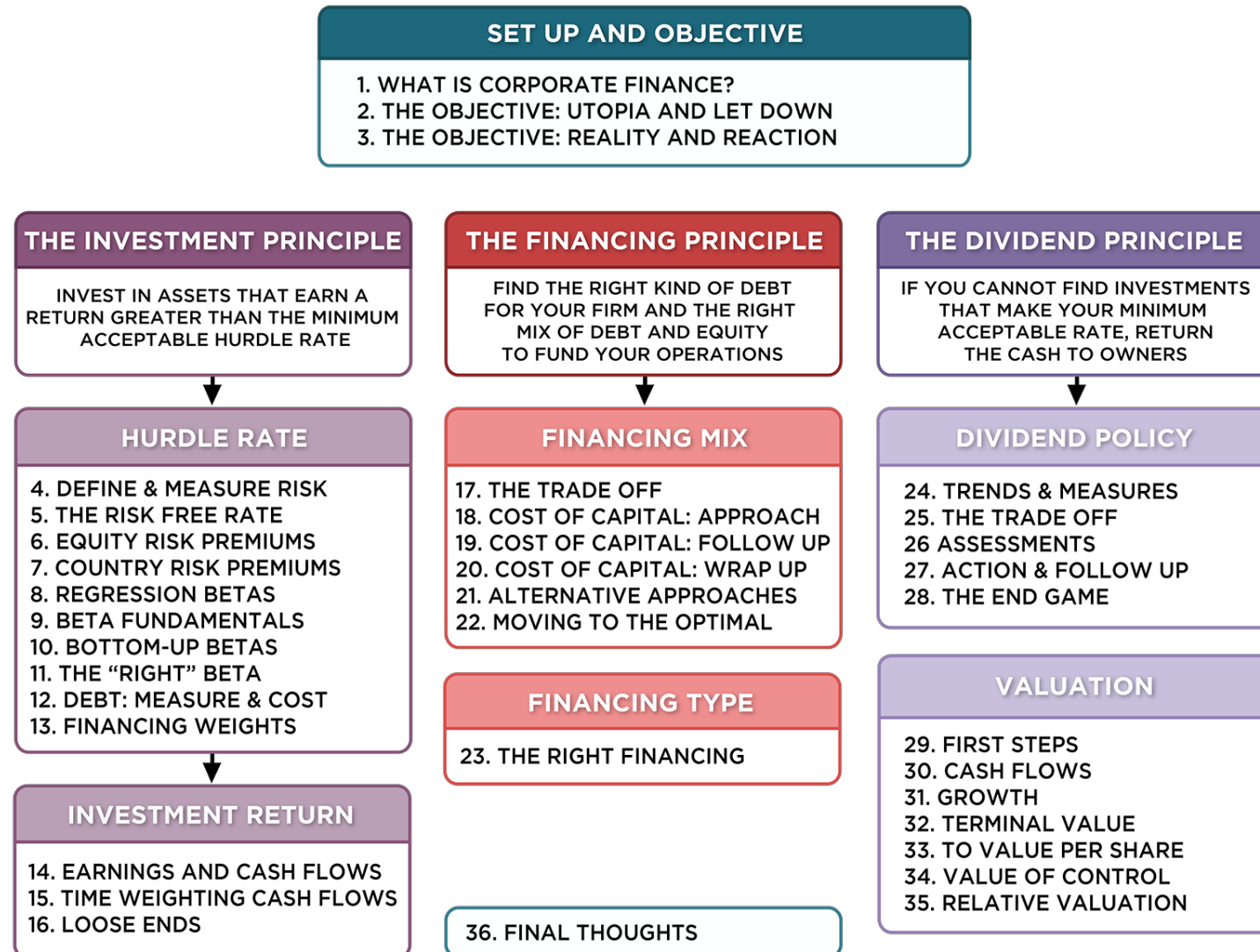


SESSION 26: ASSESSMENTS

Dividend policy rests on management trust.

SESSION 26: ASSESSMENTS



THE CASH/TRUST ASSESSMENT

Step 1: How much could the company have paid out during the period under question?

Step 2: How much did the company actually pay out during the period in question?

Step 3: How much do I trust the management of this company with excess cash?

How well did they make investments during the period in question?

How well has my stock performed during the period in question?

HOW MUCH HAS THE COMPANY RETURNED TO STOCKHOLDERS?

- As firms increasingly use stock buybacks, we have to measure cash returned to stockholders as not only dividends but also buybacks.
- For instance, for the companies we are analyzing, the cash returned looked as follows:

	<i>Disney</i>		<i>Vale</i>		<i>Tata Motors</i>		<i>Baidu</i>		<i>Deutsche Bank</i>	
Year	Dividends	Buybacks	Dividends	Buybacks	Dividends	Buybacks	Dividends	Buybacks	Dividends	Buybacks
2008	\$648	\$648	\$2,993	\$741	7,595₹	0₹	¥0	¥0	2,274 €	0 €
2009	\$653	\$2,669	\$2,771	\$9	3,496₹	0₹	¥0	¥0	309 €	0 €
2010	\$756	\$4,993	\$3,037	\$1,930	10,195₹	0₹	¥0	¥0	465 €	0 €
2011	\$1,076	\$3,015	\$9,062	\$3,051	15,031₹	0₹	¥0	¥0	691 €	0 €
2012	\$1,324	\$4,087	\$6,006	\$0	15,088₹	970₹	¥0	¥0	689 €	0 €
2008-12	\$4,457	\$15,412	\$23,869	\$5,731	51,405₹	970₹	¥0	¥0	¥4,428	¥0

A MEASURE OF HOW MUCH A COMPANY COULD HAVE AFFORDED TO PAY OUT: FCFE

- The Free Cash Flow to Equity (FCFE) is a measure of how much cash is left in the business after non-equity claimholders (debt and preferred stock) have been paid, and after any reinvestment needed to sustain the firm's assets and future growth.
 - + Depreciation & Amortization
 - = Cash flows from Operations to Equity Investors
 - Preferred Dividends
 - Capital Expenditures
 - Working Capital Needs
 - Principal Repayments
 - + Proceeds from New Debt Issues
 - = Free Cash flow to Equity

DISNEY'S FCFE: 2008-2012

	2012	2011	2010	2009	2008	Aggregate
Net Income	\$6,136	\$5,682	\$4,807	\$3,963	\$3,307	\$23,895
- (Cap. Exp - Depr)	\$604	\$1,797	\$1,718	\$397	\$122	\$4,638
- Δ Working Capital	(\$133)	\$940	\$950	\$308	(\$109)	\$1,956
Free CF to Equity (pre-debt)	\$5,665	\$2,945	\$2,139	\$3,258	\$3,294	\$17,301
+ Net Debt Issued	\$1,881	\$4,246	\$2,743	\$1,190	(\$235)	\$9,825
= Free CF to Equity (actual debt)	\$7,546	\$7,191	\$4,882	\$4,448	\$3,059	\$27,126
Free CF to Equity (target debt ratio)	\$5,720	\$3,262	\$2,448	\$3,340	\$3,296	\$18,065
Dividends	\$1,324	\$1,076	\$756	\$653	\$648	\$4,457
Dividends + Buybacks	\$5,411	\$4,091	\$5,749	\$3,322	\$1,296	\$19,869

Disney returned about \$1.5 billion more than the \$18.1 billion it had available as FCFE with a normalized debt ratio of 11.58% (its current debt ratio).

ESTIMATING FCFE WHEN LEVERAGE IS STABLE

Net Income

$$\begin{aligned} & - (1 - DR) (\text{Capital Expenditures} - \text{Depreciation}) \\ & - (1 - DR) \text{Working Capital Needs} \\ & = \text{Free Cash Flow to Equity} \end{aligned}$$

- $DR = \text{Debt/Capital Ratio}$
- For this firm,
 - $\text{Proceeds from new debt issues} = \text{Principal Repayments} + d (\text{Capital Expenditures} - \text{Depreciation} + \text{Working Capital Needs})$
 - Thus, whatever debt has to be repaid gets paid off with new debt and additional debt is taken on to fund growth in the firm.

AN EXAMPLE: FCFE CALCULATION

- Consider the following inputs for Microsoft in 1996. In 1996, Microsoft's FCFE was:
 - Net Income = \$2,176 million
 - Capital Expenditures = \$494 million
 - Depreciation = \$480 million
 - Increase in Non-Cash Working Capital = \$35 million
 - Debt Ratio = 0%
- $$\begin{aligned}\text{FCFE} &= \text{Net Income} - (\text{Cap ex} - \text{Depr}) (1 - \text{DR}) - \text{Chg WC} (1 - \text{DR}) \\ &= \$2,176 - (494 - 480) (1 - 0) - \$35 (1 - 0) \\ &= \$2,127 \text{ million}\end{aligned}$$
- By this estimation, Microsoft could have paid \$2,127 million in dividends/stock buybacks in 1996. They paid no dividends and bought back no stock. Where will the \$2,127 million show up in Microsoft's balance sheet?

FCFE FOR A BANK?

- We redefine reinvestment as investment in regulatory capital

$$FCFE_{\text{Bank}} = \text{Net Income} - \text{Increase in Regulatory Capital (Book Equity)}$$

- Consider a bank with \$10 billion in outstanding loans and book equity of \$750 million. If it maintains its capital ratio of 7.5%, intends to grow its loan base by 10% to \$11 billion and expects to generate \$150 million in net income:

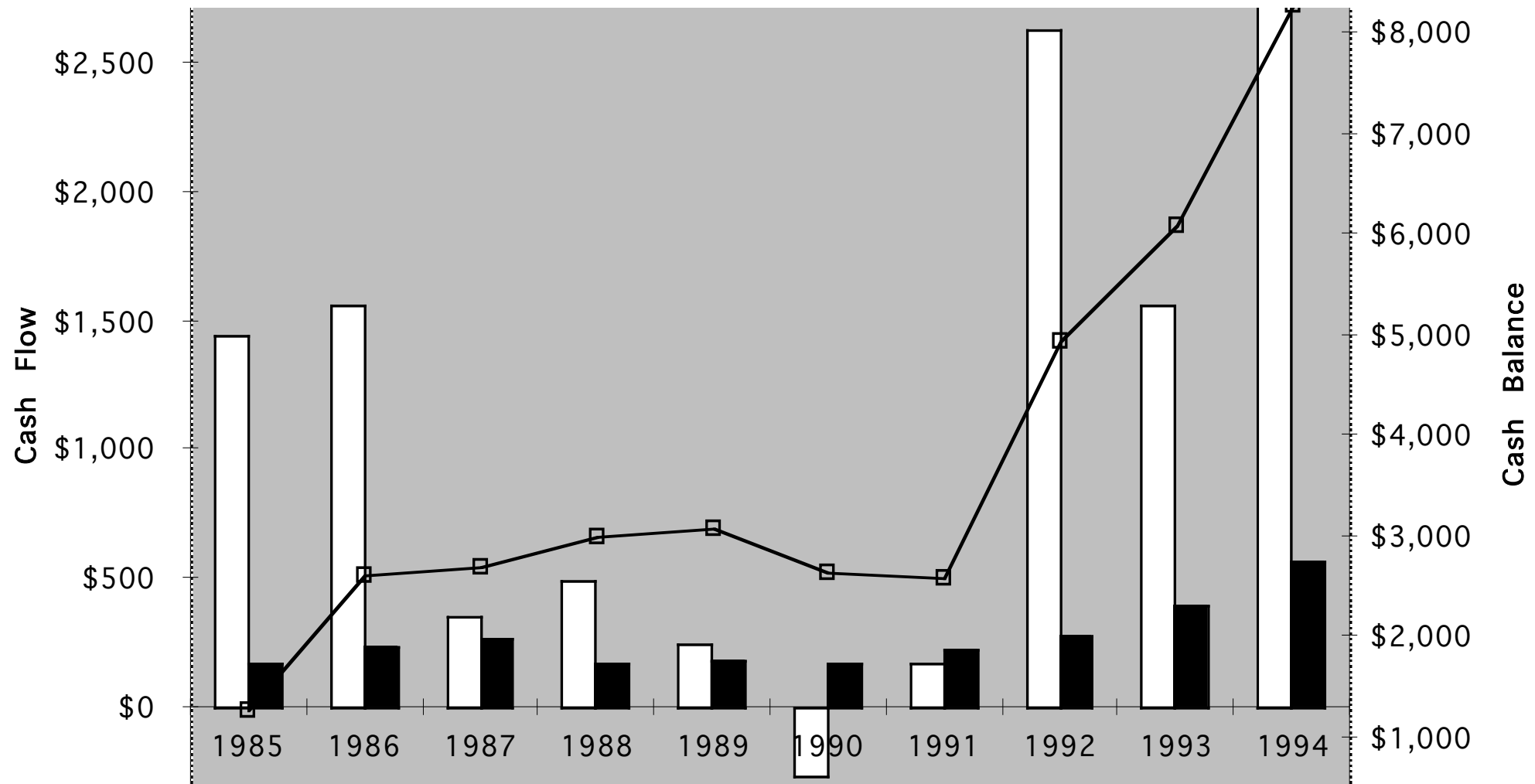
$$FCFE = \$150 \text{ million} - (11,000 - 10,000) * (0.075) = \$75 \text{ million}$$

	Current	1	2	3	4	5
Asset Base	439,851 €	453,047 €	466,638 €	480,637 €	495,056 €	509,908 €
Capital ratio	16.00%	16.00%	16.00%	16.00%	16.00%	16.00%
Tier 1 Capital	70,376 €	72,487 €	74,662 €	76,902 €	79,209 €	81,585 €
Change in regulatory capital		2,111 €	2,175 €	2,240 €	2,307 €	2,376 €
Book Equity	76,829 €	78,940 €	81,115 €	83,355 €	85,662 €	88,038 €
ROE	-1.08%	0.74%	2.55%	4.37%	6.18%	8.00%
Net Income	-757 €	584 €	2,072 €	3,642 €	5,298 €	7,043 €
- Investment in Regulatory Capital		2,111 €	2,175 €	2,240 €	2,307 €	2,376 €
FCFE		-1,528 €	-102 €	1,403 €	2,991 €	4,667 €

DIVIDENDS VERSUS FCFE: ACROSS THE GLOBE

Dividend Class	US	Europe	Japan	Emerging Markets	Aus, NZ & Canada	Global
FCFE<0, No Dividends/Buybacks	28.31%	28.38%	10.90%	21.78%	59.49%	26.91%
FCFE >0, No Dividends/Buybacks	29.86%	19.56%	13.26%	23.01%	15.76%	22.05%
FCFE >0, FCFE>Dividends+Buybacks	14.52%	22.93%	35.04%	22.98%	9.02%	21.10%
CASH ACCUMULATORS	44.38%	42.49%	48.30%	45.98%	24.77%	43.16%
FCFE >0, FCFE<Dividends+Buybacks	8.80%	9.74%	8.40%	7.91%	4.62%	8.05%
FCFE<0, 've Dividends+Buybacks	18.51%	19.38%	32.40%	24.34%	11.11%	21.88%
OVER PAYERS	27.31%	29.12%	40.80%	32.24%	15.73%	29.93%

CONSEQUENCES OF FAILING TO PAY FCFE





APPLICATION TEST: ESTIMATING YOUR FIRM'S FCFE

- In General,
Net Income
+ Depreciation & Amortization
- Capital Expenditures
- Change in Non-Cash Working Capital
- Preferred Dividend
- Principal Repaid
+ New Debt Issued

= FCFE
- Compare to
Dividends (Common)
+ Stock Buybacks

If cash flow statement used
Net Income
+ Depreciation & Amortization
+ Capital Expenditures
+ Changes in Non-cash WC
+ Preferred Dividend
+ Increase in LT Borrowing
+ Decrease in LT Borrowing
+ Change in ST Borrowing
= FCFE

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Task

Estimate the potential dividends for your company and its current cash balance.

FOURTH EDITION

Applied Corporate Finance

ASWATH DAMODARAN



WILEY

Optional:
Read
Chapter 11